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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. Note: You may be required to provide proof of your outreach to non-contributing members upon request.
N/A

STEP 1

MUTUAL FUND: Fidelity Select Technology Portfolio

(PREPARED BY: MAURICE OSAZUA OGEDEGBE)

My choice of mutual fund was Fidelity Select Technology Portfolio (FSPTX).

1. Performance

   In terms of performance, FSPTX has a 5-year annualized total return of 24.92% and a 3-year annualized total return of 7.15%. Its performance is good compared to the general market trend.

2. Fees

   Management fee charged to holder annually: 0.3%

   Management fee charged for transactions: 0.53%

   Net Expense Ratio: Annual Report: 0.69%

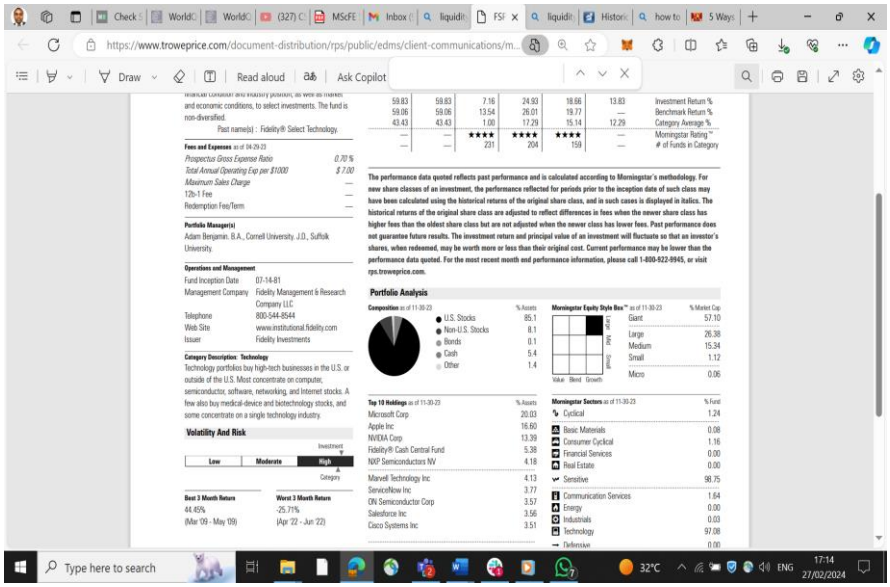
   Net Expense Ratio: Prospectus: 0.7%

3. Transparency

   FSPTX is regulated by the Securities and Exchange Commission (SEC) in the United State of America and as such publishes their financial details and other activities as expected. This confers a central regulation on the company. In addition, the fund offers dividends and capital gains twice a year in month of April and December.

4. Liquidity

   Mutual funds are generally liquid assets as fractions of funds are as cash, in cash related investment and the bulk of the funds are channeled into stocks which are liquid as well. For FSPTX, this is as shown below:



Source : FSPTX.pdf (troweprice.com)

5. Professional Management

   A history of managerial instability at FSPTX is worrisome, but things are looking good with the lead manager (Adam Benjamin) assuming the role 2014 and has been following the Technology sector for over 15 years. He is also a co-manager of Semiconductors Portfolio, which he has managed since March 2020 (Yahoo Finance).

6. Investor Protections
   The U.S. Securities and Exchange Commission (SEC) regulates the activities of FSPTX. This guarantees investor protections.

EXCHANGE-TRADED FUND: SPDR S&P 500 ETF Trust(SPY)

(PREPARED BY: XIYUAN LI)

The Exchange-traded Fund that tracks the same broad market index in this case is SPDR S&P 500 ETF Trust (SPY)

1. Performance

   Quite well, and details listed in two tables in the beneath.

Fund Characteristics as of Feb 23 2024

Est. 3-5 Year EPS Growth	**13.74%**
Number of Holdings	**503**
Price/Book Ratio	**4.44**
Price/Earnings Ratio FY1	**21.26**
Weighted Average Market Cap	**$794,662.25 M**

	As Of	1 Month	QTD	YTD	1 Year	3 Year	5 Year	10 Year	Since Inception Jan 22 1993
Fund Before Tax									
NAV	Jan 31 2024	1.67%	1.67%	1.67%	20.65%	10.84%	14.14%	12.47%	10.06%
Market Value	Jan 31 2024	1.59%	1.59%	1.59%	20.60%	10.85%	14.13%	12.47%	10.06%
Benchmark	Jan 31 2024	1.68%	1.68%	1.68%	20.82%	10.99%	14.30%	12.62%	10.19%
Fund After Tax									
Return after Taxes on Distributions	Jan 31 2024	1.67%	1.67%	1.67%	20.21%	10.45%	13.70%	11.98%	9.53%
Return after Taxes on Distributions and Sale of Fund Shares	Jan 31 2024	0.99%	0.99%	0.99%	12.47%	8.40%	11.28%	10.27%	8.78%
Benchmark	Jan 31 2024	1.68%	1.68%	1.68%	20.82%	10.99%	14.30%	12.62%	10.19%

2. Fees

   0.000945

3. Transparency

Asset details are disclosed every day, this shows high transparency. Also, ETFs are often subject to independent audits by external auditors. These audits help ensure that the fund's financial statements and operations are accurately presented and comply with accounting standards.

4. Liquidity

SPY is continuous trading throughout the day on the stock exchange, so it has a good liquidity

5. Professional Management

The trustee is "State Street Global Advisors Trust Company", one of the biggest liquid ETFs globally. The fund is managed by passionate and experienced manager, that shows high trustworthiness.

6. Investor Protections

The U.S. Securities and Exchange Commission (SEC) is the primary regulatory body overseeing SPY and other investment funds. ETFs, as registered investment companies, must comply with SEC regulations. The SEC reviews and approves the registration statements and ongoing filings of ETFs to ensure disclosure and regulatory requirements are met.

INDEXED ANNUITY: Athena Indexed Annuity

(PREPARED BY: MIN CHEUNG)

1. Performance

The Athena Indexed Annuity aims to provide a balance between growth potential and protection against market downturns. Returns are linked to a specific market index, with a portion of the index's gains credited to the annuity. The exact performance metrics and historical returns should be detailed in accompanying tables or documents.

2. Fees

Indexed annuities often come with a range of fees, including administrative fees, mortality, and expense risk charges, and potentially surrender charges for early withdrawal. The specific fee structure for the Athena Indexed Annuity, including any caps or participation rates, should be clearly outlined.

3. Transparency

The terms and conditions of the Athena Indexed Annuity, including how interest is credited and any caps or participation rates, are disclosed upfront. This product may also include statements or reports that detail the annuity's performance and the crediting method used to calculate returns.

4. Liquidity

Indexed annuities typically offer limited liquidity options, such as free withdrawals of a certain percentage of the account value annually. The specific liquidity features of the Athena Indexed Annuity, including any penalties for early withdrawal, should be detailed.

5. Professional Management

The Athena Indexed Annuity is managed by experienced professionals who specialize in annuity products and investment strategies. The management team's expertise in balancing risk and return to meet the annuity's objectives should be highlighted.

6. Investor Protections

Indexed annuities are insurance products and are regulated by state insurance commissions rather than the SEC. The Athena Indexed Annuity provides certain guarantees, such as a minimum interest rate and return of premium, which offer a level of protection to investors.

STEP 2

7. Collateral-Related Risks. Credit Risk

MUTUAL FUND: Fidelity Select Technology Portfolio

a. Mutual funds are prone to credit risk in terms of market risk as they are subject to large volatility and as such in a time of economic downturn, their net asset value goes low. There is also the risk of financial impropriety on the part of the fund manager which is an important risk in investing in mutual funds. Finally, mutual funds are exposed to the direct risks of companies that owns the assets that form bulk portion of the mutual fund’s portfolio.

b. The Security and Exchange Commission (SEC) is naturally saddled with the responsibility of insuring the payment of returns and grantee credit guarantee to investors in mutual funds.

c. How is credit guarantee appropriated for annuity investors in a period of financial crisis?

EXCHANGE-TRADED FUND: SPDR S&P 500 ETF Trust(SPY)

a. ETFs, especially SPY（SPDR S&P 500 ETF Trust）, do not directly hold securities, so its

risk do not include default. It mainly influenced by the financial performance for S&P 500 companies, and the integral market situation. Besides, the risk of publisher's bad management is also matter.

b. Firstly, is its publisher State Street Global Advisors that has the responsibility, and then the supervisors like FINRA and SEC has obligation to ensure the normal operation of ETF. And last is SEC, which is the primary regulatory body overseeing SPY and other investment funds. The SEC reviews and approves the registration statements and ongoing filings of ETFs to ensure disclosure and regulatory requirements are met.

c. Is there any protection measure to prevent huge loss?

INDEXED ANNUITY: Athena Indexed Annuity

a. The primary credit risk in an indexed annuity is related to the issuing insurance company's ability to fulfill its contractual obligations. If the insurer faces financial difficulties, it may impact the guarantees provided by the annuity.

b. The guarantees in indexed annuities are backed by the issuing insurance company. Additionally, state guaranty associations provide a safety net, albeit with limits, in case the insurer defaults.

c. What are the financial stability and credit ratings of the insurance company issuing the annuity? How do state guaranty associations protect annuity holders?

8. Statistical-Related Risks: Correlation.

MUTUAL FUND: Fidelity Select Technology Portfolio

a. High correlation signals low diversity which as well is a precursor to risk in a normal market. So medium to high correlation might be of negative importance in investment safety. But correlation alone as a factor should not be considered singly without looking at risk tolerance, investment goal and duration of investment. Another point to note is that medium to high correlation in themselves shows that assets may not outperform the general market and thus returns on investments may be within the range of inflation.

b. With a higher risk of default in EFT and mutual funds during a period of crisis as compared to annuity, what is the haven for investors?

c. What is the correlation like between participation rate and the equity market performance in periods of market downturn on the one hand and when the market is bullish?

EXCHANGE-TRADED FUND: SPDR S&P 500 ETF Trust(SPY)

a. No, because high correlation means it is not diversified, and one break will cause the whole collapse.

b. Will these instruments be influenced by down markets?

c. Is the relationship between participation rate and the equity market performance negative correlated

or positive correlated?

INDEXED ANNUITY: Athena Indexed Annuity

a. Indexed annuities are designed to provide a level of protection against market downturns due to their downside protection feature. Even if the equities in the index are highly correlated and experience a downturn, the annuity's principal is typically protected.

b. How does the ETF manage risks associated with high correlation in down markets? What strategies are in place to mitigate losses during market volatility?

9. **Magnifying Risk: Leverage & Nonlinearity**

MUTUAL FUND: Fidelity Select Technology Portfolio

a. Indexed annuity by design appeals to investors want some exposure to the equity market but also want to prevent losses. This guarantees a financial protection against economic downsides. The guaranteed interest rate range also underscores this fact.

b. Since indexed annuity provide a balance between growth and safety, aside from offering downside protection as stated in (a) above, they also allow for possibility for investment growth, offer a guaranteed minimum interest rate, and earnings are allowed to grow tax deferred as long as they within the annuity.

c. The participation rate determines the percentage of the index return applicable as interest credit for investor. The higher the participation rate, the higher the investment leverage. So, investors that are more risk taking would adopt a higher participation rate.

d. There is a positive correlation between the participation and equity market performance is positive. A good performance in the underlying stock propels an increased rate of participation by investors in indexed annuities.

EXCHANGE-TRADED FUND: SPDR S&P 500 ETF Trust(SPY)

An indexed annuity typically has a guaranteed minimum interest rate that resets annually. This means that the interest rate credited to the annuity is adjusted annually to ensure a minimum guaranteed return. While this provides a degree of stability, the annual reset may introduce certain features that investors should be aware of.

Over time, the compounding effect of the Guaranteed Minimum Interest Rate may result in the accumulation of value within the index annuity. However, annual resets can also affect overall performance. If the underlying index experiences

significant volatility, annual resets may affect the potential for magnified returns or losses.

As a result of the annual reset mechanism, the long-term performance of an index annuity may deviate from the expected growth in returns based on the underlying index. This deviation is a result of the annually compounded interest rate and may have a differential impact on the overall long-term performance of the annuity.

The Guaranteed Minimum Rate provides a degree of stability, but the reset mechanism may introduce a degree of volatility into the performance of the annuity. Annual adjustments may result in fluctuations in credited interest, affecting the overall volatility of the annuity. This can make annuities more sensitive to changes in market conditions, particularly during periods of economic uncertainty.

INDEXED ANNUITY: Athena Indexed Annuity

a. Indexed annuities offer a guarantee that the principal investment will not decrease due to market downturns, providing a safety net for the investor's initial investment.
b. The upside in indexed annuities comes from the participation rate, which determines how much of the index's gain is credited to the annuity. This allows for growth potential while maintaining principal protection.
c. The participation rate acts as a form of leverage by determining the percentage of the index's gains that will be credited to the annuity. A higher participation rate means higher potential returns, but it's important to understand how this rate is set and adjusted over time, as it affects the annuity's growth potential.
d. The correlation between the participation rate and equity market performance is crucial. In strong markets, a high participation rate can significantly enhance returns, but if the rate is adjusted downward in response to market conditions, it can limit the upside potential.

STEP 3

10. Frictional Factors: Liquidity and Regulation

a. MUTUAL FUND: Fidelity Select Technology Portfolio

i. Mutual fund has a relatively secured outlook compared to ETF and other volatile assets. The same reason why it has more funds from the pension fund space invested in it. In addition, these funds allow for both short term and long-term categories of investment which accounts for the large volume of inward and outward transactions. It should be noted that this pattern does not speak to the performance of this class of product/asset.

ii. Mutual funds do not trade intraday for their peculiar attribute as related to calculations of their net asset value only daily (end of day), despite the incessant price fluctuations. This then allows investors access to trade at the end of day regardless of when trade orders were placed. In addition, trade decisions are made only by fund managers throughout the day based on their investment strategy and portfolio combination which makes intraday trading not realistic.

iii. Despite disclosure requirements from regulatory agencies, certain important details may be left out as they may not be statutory but important for making investment decisions. The time lag of 60 days allowable period before quarterly filing with SEC might be a cause for concern for the public. Finally, mutual funds normally disclose only their top 10 assets thus allowing shrouding information on other assets.

b. EXCHANGE-TRADED FUND: SPDR S&P 500 ETF Trust(SPY)

i. SPY is exposed to market risk, and during market crises, the fund's value may decline significantly, reflecting broader market conditions. While the S&P 500 Index represents a broad and diversified group of large-cap stocks, it is not immune to market downturns.

ii. Gross Expense Ratio: 0.000945.

iii. Dividends affect the SPDR S&P 500 ETF Trust (SPY). SPY is an exchange-traded fund that tracks the performance of the S&P 500 Index, which includes large-cap U.S. equities. Dividends on individual stocks held by SPY play a role in affecting the fund's net asset value (NAV), which represents the total value of the fund's assets minus liabilities divided by the

number of shares outstanding SPY receives income when stocks in the S&P 500 Index pay dividends. This dividend income has several effects on the Fund.

11. REGULATION

MUTUAL FUND

Part 1: Crisis

In the American mutual fund market, situations such as illegal/unethical trading, late trading, market timing, illegal sales practices amongst other unethical practices led to the financial abuse/scandal and subsequent crisis. The crisis was more pronounced and huger as the big players in American finance were involved. The full weight of the effect of the crisis was avoided by investigating the mutual fund industry in relation to market timing and late trading. The damages of the crisis were estimated to be over US $4 billion per year.

Despite the creation of the Securities and Exchange Commission and the Investment Company Act of 1940, by 2023, late trading and market timing for selected and 'weighty' clients resurfaced.

Information asymmetry and deliberate late trading allowing trade take place beyond the cut off time were predominantly the issue at hand. The problem at hand was causing a negative effect on long term investments and eroding market confidence. Illegal compensation paid to mutual fund companies was also another major factor that sustained this illegality. This also drove other investors to trade in mutual funds causing a false market growth.

Despite standing regulations around the heinous act of the mutual fund companies, the issue became compounded as monitoring market timing and enforcement became the prerogative of each mutual fund company.

In addition, with the large influx of investors, funds into the mutual fund market especially with the advent of the individual retirement accounts, there was a large pool of funds to allow for skimming.

The crisis eventually thrived on the wings of lack of transparency caused by a shadow cast in the operation of the brokerage firms and mutual fund companies such that the mutual funds were handicapped.

Another major enabling factor for the crisis was traced to unhealthy competition in mutual fund industry allowing for fraudulent.

There was also the challenge posed by conflict of interest mainly because actors were not independent and thereby had no responsibility protecting shareholders' interest thus allowing for fraudulent tendencies within their organization.

Part 2: Proposed Regulation

Initially, there was the creation of the Securities and Exchange Commission (SEC) to oversee the industry in the American market. Later on, the Investment Company Act of 1940 was brought to regulate the mutual fund industry by requiring all funds to register with the SEC and meet standards of operation, disclosure, and securities transactions, which had been absent until this period. Under this Act, "backward pricing," which was a loophole was corrected in 1968 when Rule 22c-1 was adopted by the SEC, thus making forward pricing the standard pricing method.

Even though the crisis did not precede the existing laws and regulations, centralized enforcement was the issue here as monitoring market timing and enforcement was the prerogative of each mutual fund.

Since the American mutual fund market was challenged with lack of transparency, the SEC has attempted to address the lack of transparency through regulation further regulation which allowed mutual funds to obtain more information about investors' transactions as well as removing brokerage firms' ability to use omnibus accounts for mutual fund investments.

To address the issue around unhealthy competition hinged on making unwholesome commissions, brokerage commissions were deregulated by 1 May 1975 and the fixed rates that had existed since the stock market's beginning were eliminated.

Furthermore, the SEC also determined to make a move allowing only 75% of the directors of the board also be independent. So, now, with the 1940 Securities Act, only 40% of the directors of the board are independent from the fund management company.

The journal also delves into outlook in Great Britain that passed the Financial Services Act of 1986, referred to as the "Big Bang." This legislation has removed fixed commissions among London's securities market so that competition could lower commissions and, therefore, reduce prices for consumers in a bid to reclaim the business opportunities lost to the NYSE after the United States made this move earlier.

Considering Japan's outlook and response to the adoption and localization of the Big Bang deregulation, there was a negative result as this led to a crisis. Lack of transparency and information asymmetry were the major challenges with Japanese version of the American mutual fund. The peculiar business culture in Japan is also not a helper factor to allow for whistleblowing which is a formidable way to attract regulatory attention to fraud perpetration. This thus limits the ability to recognize crises and re-regulate when scandals are discovered unlike their American and Great Britain counterparts. Japanese investment strategy which is aimed at benefitting from monetary insurance through investor not focusing solely on financial objectives is also sub optimal to a thriving regulation in the investment trust. However, a great boost to Japanese investment trust is the culture to deregulate slowly thus allowing for proper monitoring of the market.

EXCHANGE-TRADED FUND

Part 1: Crisis

Synthetic ETFs face specific challenges and risks in times of crisis, which are critical for investors considering these funds. The following is a detailed analysis of the potential crisis for synthetic ETFs:

1. **Counterparty Risk:**

Synthetic ETFs rely on derivative contracts, such as total return swaps, with counterparties (usually banks). Counterparty risk becomes more pronounced during a financial crisis or recession. If a counterparty faces financial difficulties or defaults, this may affect the ETF's ability to replicate index returns. Investors may face losses if the counterparty fails to fulfill its obligations. The financial condition and creditworthiness of the counterparty becomes a key factor.

2. **Market liquidity and pricing challenges:**

In times of crisis, market liquidity may be reduced, affecting the pricing of ETFs and the derivatives they hold. Synthetic ETFs may experience wider bid-ask spreads, making it more challenging to execute trades at favorable prices. Investors may find it more difficult to buy and sell shares at fair value, which may result in increased transaction costs and differences between the market price of the ETF and its net asset value (NAV).

3. **Regulatory Review:**

Regulators may increase their scrutiny of the use of derivatives, particularly during periods of market stress. Changes in the regulatory environment may affect the operational efficiency and viability of synthetic ETFs. More stringent regulation may affect a synthetic ETF's ability to use certain derivatives or increase compliance costs, thereby affecting the overall structure and fees of the fund.

4. **Tracking error amplification:**

Synthetic ETFs aim to replicate the returns of an index through derivatives. During periods of market volatility or disruption, tracking error (the deviation between the ETF's performance and the index) may increase, resulting in poor tracking. Investors may receive returns that are significantly different from the expected index performance, thereby affecting the fund's ability to achieve its investment objective.

5. liquidity **mismatch:**

The underlying assets of synthetic ETFs may have different liquidity profiles compared to the derivatives they use. In times of crisis, the liquidity of the derivatives market and the underlying securities may be inconsistent, which may affect the ETF's ability to manage its portfolio effectively. Such mismatch may create challenges for redemptions and put liquidity pressure on the fund.

6. **Operational complexity:**

The operational structures of synthetic ETFs involve complex arrangements with counterparties and derivatives. Managing the complexity of these structures becomes challenging during a crisis. Increased operational risk and potential disruptions to the execution of derivative strategies may affect the fund's ability to deliver index-consistent returns.

7. **investor confidence:**

Market uncertainty and heightened risk during a crisis may erode investor confidence, particularly in more complex investment structures such as synthetic ETFs. Investors may become wary of synthetic ETFs, leading to increased redemption pressure. Loss of confidence may exacerbate liquidity challenges and lead to a decline in the fund's market capitalization.

Conclusion:

Synthetic ETFs can offer cost advantages, but specific risks associated with counterparty relationships, market conditions and the regulatory environment make them crisis-sensitive. Investors should carefully assess these risks, monitor counterparty financials, and stay abreast of regulatory developments. Diversification and due diligence remain critical to managing the potential challenges associated with synthetic ETFs during a crisis.

Part 2: Regulation

1. Enhancing counterparty risk management:

Involving setting stricter standards for the selection and monitoring of counterparties, ensuring transparency of counterparty exposures, and developing contingency plans in the event of a counterparty default.

2. Increased disclosure requirements:

disclosure requirements by regulators including detailed information on counterparty relationships, derivatives used and stress test results.

3. Liquidity risk management:

Regulation is likely to focus on ensuring robust liquidity risk management practices for ETFs, particularly during periods of market stress. This may involve guidelines for maintaining secondary market liquidity, managing liquidity mismatches and providing clear information about the liquidity of underlying assets and derivatives.

4. Fund structure review:

Regulators may review the structure of synthetic ETFs to assess whether changes are needed to enhance investor protection and market stability. This may involve consultation with industry stakeholders to gather insights on potential improvements to the fund structure.

5. Cooperation with international bodies:

Given the global nature of financial markets, regulators could work with international bodies to develop consistent standards and guidance for synthetic ETFs. This may involve coordination with entities such as the International Organization of Securities Commissions (IOSCO).

INDEXED ANNUITY

Crisis

In 2003, a scandal erupted in the US mutual fund industry when it was discovered that several prominent mutual fund companies had engaged in illegal trading practices that harmed average investors. The main abusive practices were: Late Trading This involved allowing certain privileged investors to place orders to buy or sell mutual fund shares after the 4 pm market close while still getting that day's price. This allowed them to profit from news events that occurred after market close. This violated forward pricing rules that require trades to be conducted at the next calculated price after the order is placed. Market Timing This involves rapidly trading in and out of mutual fund shares to exploit short-term price fluctuations. While not illegal per se, most mutual funds claimed in their prospectuses that they did not allow market timing. Yet fund companies were found to be allowing and even encouraging this practice for favored clients, to the detriment of long-term investors. The crisis uncovered conflicts of interest and lack of transparency that

allowed such abuses to occur unchecked. Estimates suggest these practices cost investors over $4 billion per year.

Proposed Regulations

In response, regulators proposed several rule changes: 1) Strengthen fund governance - Require an independent fund chairman and 75% independent directors to better represent shareholder interests over fund companies. 2) Increase transparency - End omnibus accounts that hide client identities from funds, implement electronic trading platforms to prevent manual tampering. Require disclosing portfolio managers. 3) Redemption fees - Deter market timers by imposing fees specifically on short-term trading. 4) Compliance procedures - Strengthen compliance monitoring to ensure funds follow their disclosed policies. These measures aim to eliminate the conflicts of interest and lack of transparency that enabled trading abuses to proliferate while restoring integrity to mutual fund governance and protections for average investors. Strict enforcement will be key to preventing repeats of such large-scale scandals. The discrepancy between physical and synthetic ETFs relates to their replication methods. Physical ETFs hold the actual securities in the index they track. Synthetic ETFs use derivatives like swaps to mimic index performance. Synthetics have faced stigma over counterparty risk and transparency. However, some evidence suggests synthetics can outperform physical ETFs in certain asset classes due to lower costs. Rather than superiority of one over the other, the needs of investors based on region, tax jurisdiction etc will determine suitable structures. As for suitability, while indexed annuities offer some downside protection, they also cap upside potential. They involve complex guaranteed income riders and high surrender charges. Hence they may not be appropriate for investors with long time horizons seeking higher returns or more liquidity. Proper due diligence is required to ensure clients understand the tradeoffs before recommending indexed annuities

REFERENCES

https://www.troweprice.com/document-distribution/rps/public/edms/client-communications/morningstar-fact-sheets/FSPTX.pdf

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